

INTRODUCTION

In recent years, fitness and wellness applications have become increasingly popular as people seek healthier lifestyles and personalized health solutions. However, most traditional fitness applications follow a generalized approach and often fail to consider the unique physiological and hormonal changes experienced by women during different phases of the menstrual cycle.

Hormonal fluctuations throughout the cycle can significantly affect energy levels, mood, strength, endurance, recovery, and overall physical performance. Ignoring these changes may lead to ineffective workout routines, discomfort, fatigue, and reduced motivation. This highlights the need for a smarter and more personalized fitness solution designed specifically for women.

BloomBalance is an AI-powered cycle-aware fitness planning application developed to address this gap by providing personalized workout recommendations based on menstrual cycle phases, symptoms, energy levels, sleep patterns, and fitness goals. The application identifies the user's current cycle phase, including menstrual, follicular, ovulation, and luteal phases, and adapts workout intensity and exercise selection accordingly. By aligning workouts with the body's natural hormonal rhythms, BloomBalance aims to promote healthier, safer, and more sustainable fitness habits.

The application includes onboarding features where users can enter personal and health-related details such as age, height, weight, cycle start date, symptoms, sleep hours, workout preferences, and fitness goals. Using intelligent recommendation logic, the system generates customized workout plans containing suitable exercises, repetitions, duration, and intensity levels. The app also supports workout tracking, progress analytics, streak monitoring, calorie estimation, and workout reminders to help users maintain consistency and motivation.

BloomBalance is developed using React Native, Expo, TypeScript, and AsyncStorage, providing a responsive, modern, and user-friendly mobile experience. By combining women's health awareness with intelligent fitness planning, BloomBalance aims to deliver a smarter and more adaptive wellness solution that encourages long-term physical and mental well-being.

SOFTWARE REQUIREMENTS

To implement the proposed BloomBalance application, several software tools and technologies are used, each contributing to the development of a personalized and cycle-aware fitness planning system. These tools help in building a responsive mobile application, managing user data, generating adaptive workout plans, and improving overall user experience. The following software requirements are essential for the successful functioning of the system:

1. *VS Code*

Visual Studio Code (VS Code) is used as the integrated development environment for coding, debugging, and managing the project files. It provides extensions, debugging tools, syntax highlighting, and version control integration that simplify the development process.

2. *Git and Github*

Git and GitHub are used for version control and project management. They help in tracking code changes, collaborating with team members, maintaining backups, and hosting the project repository online.

3. *React native*

React Native is the primary framework used for developing the BloomBalance mobile application. It enables cross-platform mobile app development using a single codebase for both Android and iOS devices. React Native provides reusable components, smooth performance, and efficient UI rendering, making it suitable for developing an interactive and user-friendly fitness application.

4. *Expo*

Expo is a development platform built on top of React Native that simplifies application development, testing, and deployment. It provides built-in support for features such as notifications, device testing, and application configuration. Expo allows faster development and easy testing using the Expo Go application, making it highly effective for rapid mobile app development.

5. *Typescript*

TypeScript is used as the programming language for developing the application logic. It is a superset of JavaScript that provides static typing, improved code maintainability, and better error detection during development. TypeScript enhances code reliability and scalability, especially for larger applications with multiple components and screens.

6. *Expo Router*

Expo Router is used for navigation management within the application. It enables smooth navigation between screens such as onboarding, goal selection, workout plans, analytics, and profile management. The routing system helps organize the application structure efficiently and improves the overall user experience.

7. *AsyncStorage*

AsyncStorage is used for persistent local data storage within the application. It stores user information, workout history, completed sessions, progress tracking data, and application preferences directly on the device. This ensures that user data remains accessible even after restarting the application.

6. *React Native chart kit*

React Native Chart Kit is utilized for generating analytics and progress visualization charts. It helps users monitor workout consistency, calorie estimation, progress trends, and activity streaks through graphical representations. The charts improve user engagement by providing clear and interactive fitness insights.

7. *Expo-notifications*

Expo Notifications is used to implement local workout reminders and notifications. It helps users maintain consistency by sending scheduled reminders for workouts, progress updates, and fitness activities. This feature enhances user motivation and daily engagement with the application.

8. *React-native-animatable*

React Native Animatable is used for adding animations and smooth transitions within the application interface. It improves the visual appeal of the application through animated cards, transitions, and interactive effects, thereby enhancing the overall user experience.

9. AI Recommendation Engine

The AI Recommendation Engine is used to generate personalized workout plans based on menstrual cycle phase, symptoms, energy levels, sleep patterns, and fitness goals. The system analyzes user inputs and dynamically creates adaptive workout recommendations to improve personalization and user engagement.

IMPLEMENTATION

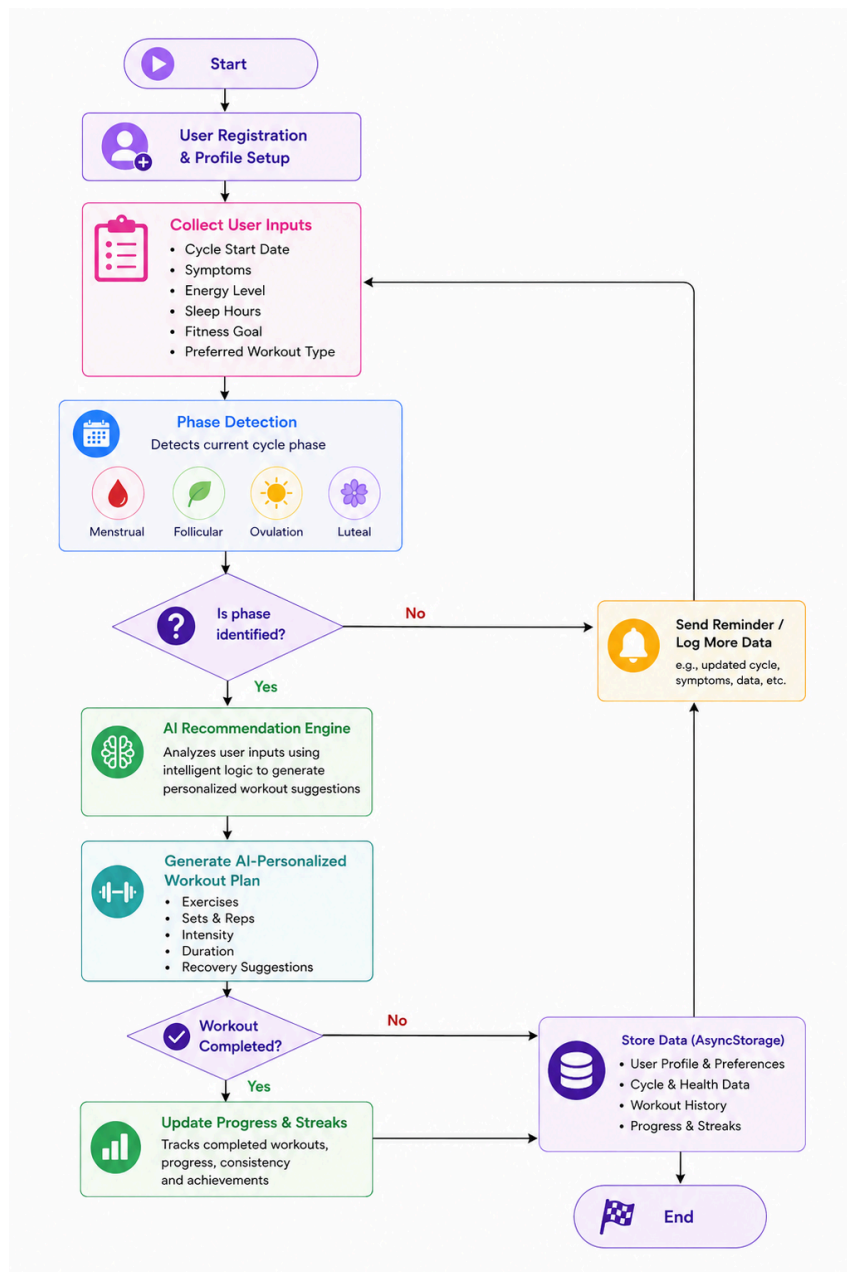


Fig 3.1 Design Diagram of our solution

SOURCE CODE (sample) :

Link: <https://github.com/nikpfsgvi8h/BloomBalanceAPP>

// Welcome Screen Logic

```
useEffect(() => {  
  
  const checkUser = async () => {  
  
    const savedUser =  
      await AsyncStorage.getItem(  
        "userData"  
      );  
  
    if (savedUser) {  
  
      router.replace("/plan");  
  
    }  
  
  };  
  
  checkUser();  
  
}, []);
```

// Save User Data

```
const userData = {  
  
  age,  
  height,  
  weight,  
  sleep,  
  energy,  
  symptoms,  
  cycleDate,  
  goal,  
  focus,  
  
};  
  
await AsyncStorage.setItem(  
  "userData",  
  JSON.stringify(userData)  
);
```

// Cycle Phase Logic

```
if (cycleDay <= 5) {  
    phase = "Menstrual Phase";  
    workout = "Recovery Yoga";  
    intensity = "Low Intensity";  
}  
else if (cycleDay <= 13) {  
    phase = "Follicular Phase";  
    workout = "Full Body Strength";  
    intensity = "Moderate Intensity";  
}  
else if (cycleDay <= 16) {  
    phase = "Ovulation Phase";  
    workout = "HIIT & Cardio";  
    intensity = "High Intensity";  
}  
else {  
    phase = "Luteal Phase";  
    workout = "Pilates Burn";  
    intensity =  
        "Low to Moderate Intensity";  
}  
  
// Energy Level Logic  
  
if (energy === "low") {  
    energyWidth = "30%";  
}  
else if (energy === "medium") {  
    energyWidth = "60%";
```

```

    }
    else if (energy === "high") {

        energyWidth = "100%";

    }

    // Workout Completion Tracking

    <TouchableOpacity
    style={styles.completeButton}

    onPress={() =>
        setStatus(
            "Workout Completed 
        )
    }
    >

    <Text style={styles.buttonText}>
        Mark as Complete
    </Text>

</TouchableOpacity>

// Logout System

const logout = async () => {

    await AsyncStorage.removeItem(
        "userData"
    );

    router.replace("/");

};

// Logo Integration

<Image
    source={require(
        "../assets/images/BloomBalancelogo.jpg"
    )}

    style={styles.logoImage}
/>

```

Note:

The above code is a part of the entire project and belongs to the file “app/index.tsx”. It is just one among the various files and hence, not the complete source code. The whole source code can be accessed from the link provided above.

INTERPRETATION OF RESULT

Fig 5.1 Home Screen

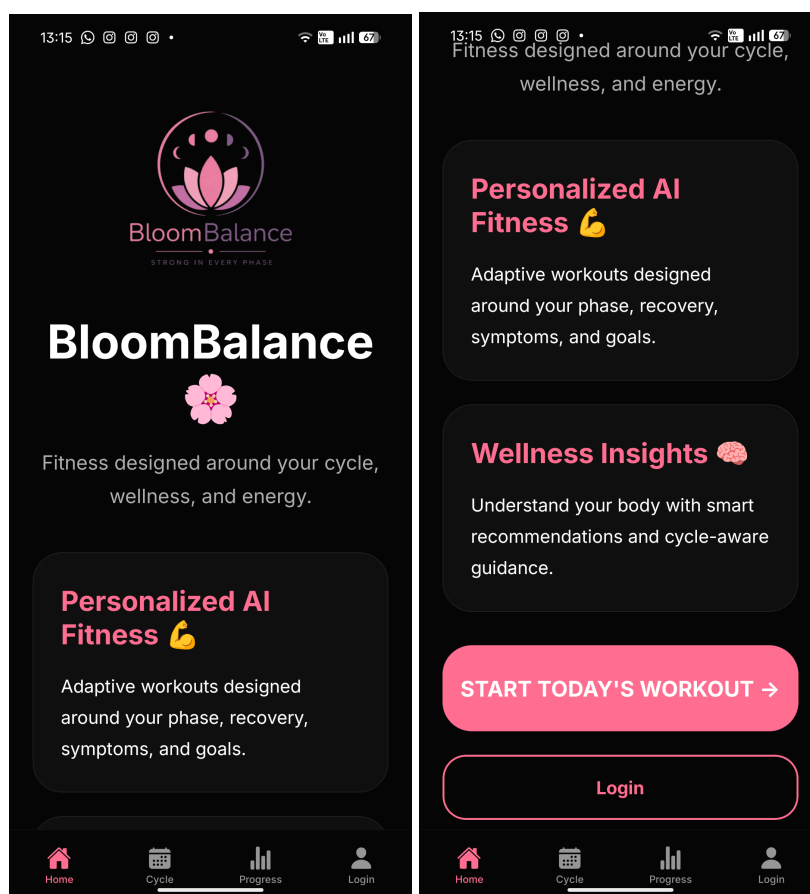


Fig 5.2 User Input screen

Create Your Wellness Profile 🌸

Personalized AI fitness designed around your cycle, energy, and goals.

Your Name

Enter your name

Age 🧑

13 14 15 16 17

Weight (kg) 📊

30 31 32 33 34

Fitness Goal 🎯

Weight Loss Strength

Wellness Muscle Gain

Focus Area 🏋️

Lower Body Upper Body Core

Full Body

Energy Level ⚡

Low Medium High

First Day of Last Period 🌸

Sat May 23 2026

Generate My AI Plan ✨

Fig 5.3 Workout Overview screen

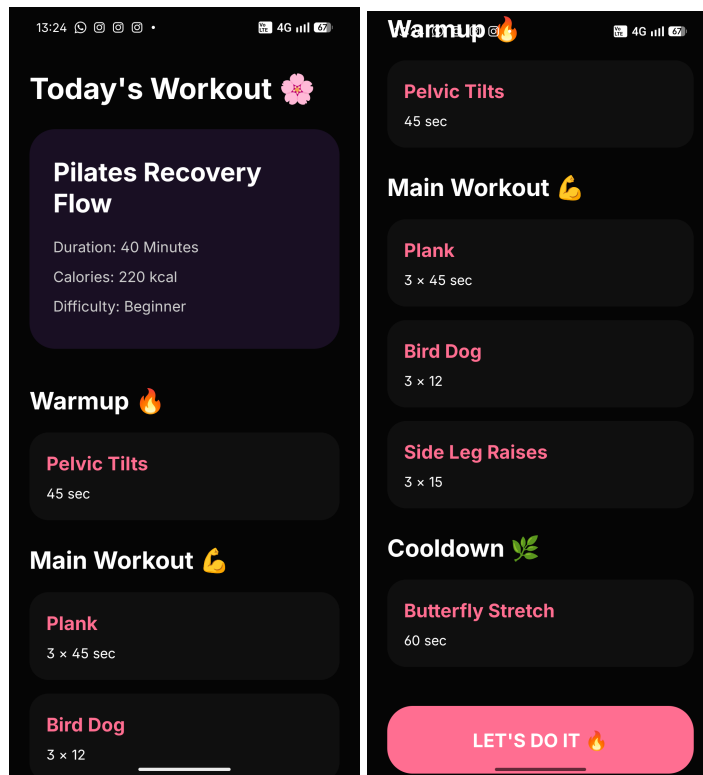


Fig 5.4 Guided Workout screen

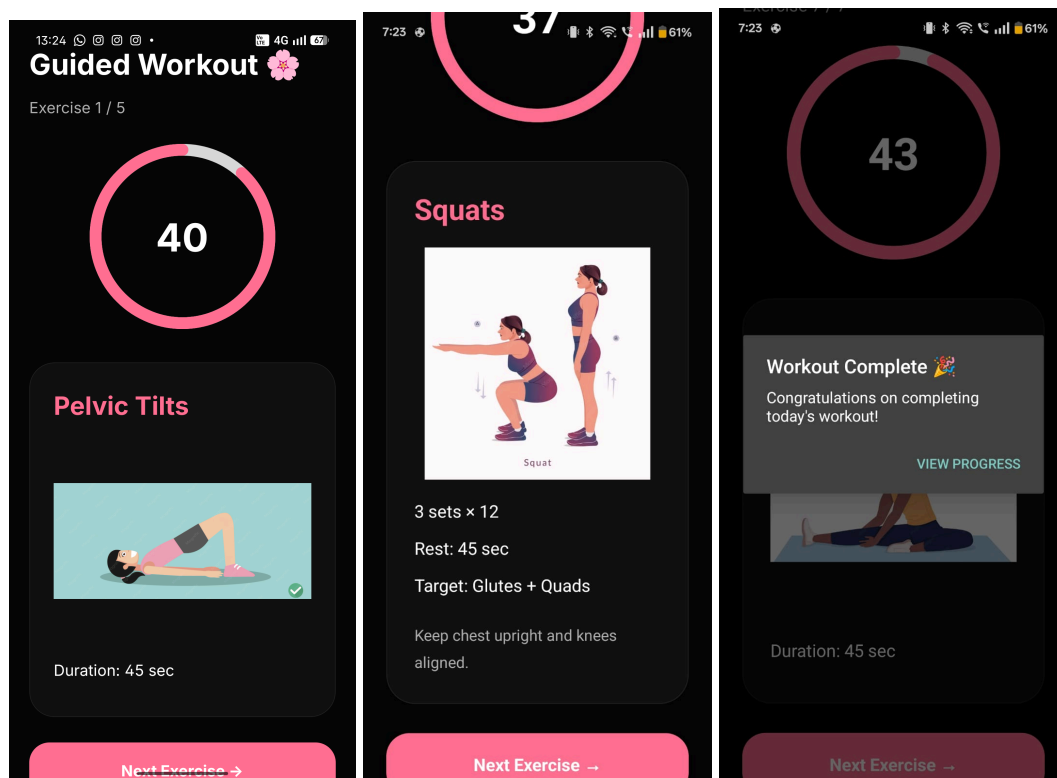


Fig 5.5 Progress screen

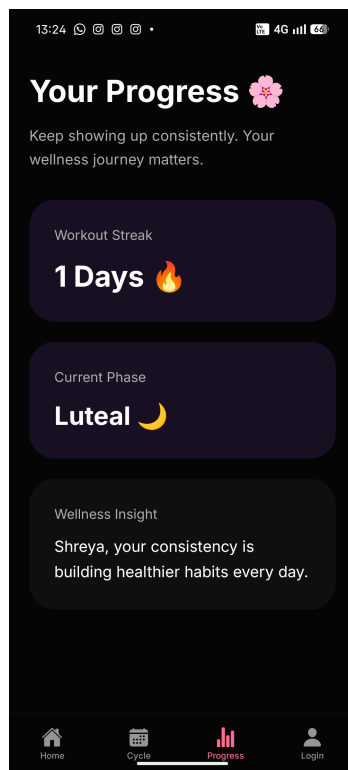
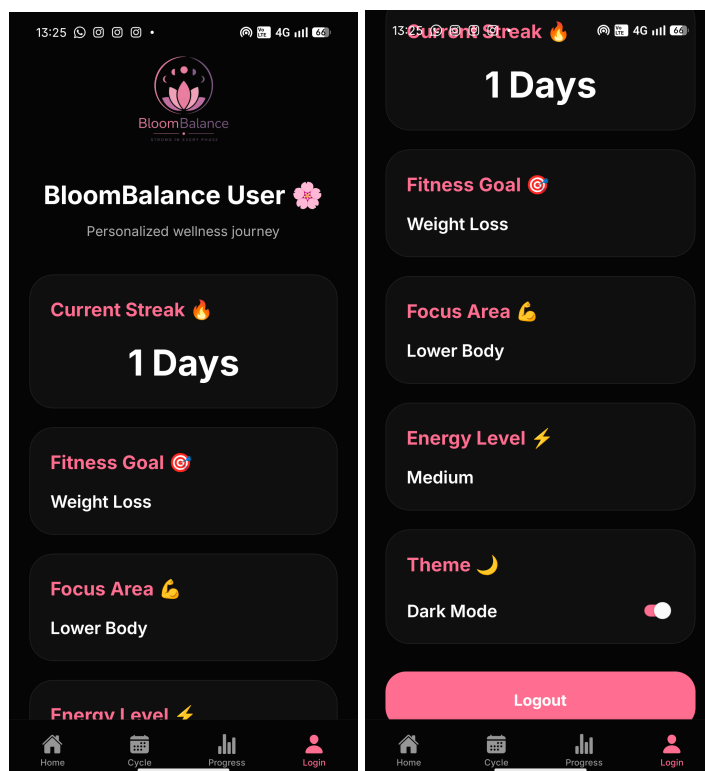


Fig 5.6 User Profile Screen



CONCLUSION AND FUTURE SCOPE

BloomBalance is an AI-powered cycle-aware fitness planning application designed to provide personalized workout experiences for women based on menstrual cycle phases, symptoms, energy levels, sleep patterns, and fitness goals. Unlike traditional fitness applications that follow a generalized approach, BloomBalance adapts workout recommendations according to the user's physical and hormonal changes throughout different phases of the menstrual cycle, helping users maintain safer and more effective fitness routines.

The application integrates cycle phase detection, AI-based workout recommendation, progress tracking, analytics visualization, local storage, and workout reminders within a user-friendly mobile interface. By combining women's wellness awareness with intelligent fitness planning, BloomBalance promotes consistency, recovery, and overall well-being.

Developed using React Native, Expo, TypeScript, and supporting libraries, BloomBalance demonstrates how intelligent mobile technology can deliver adaptive health and fitness solutions for women.

1. Nutrition and Meal Planning

The application can be expanded to include personalized nutrition and meal planning features. Based on the user's menstrual cycle phase, fitness goals, and health conditions, the app can recommend suitable diets, hydration plans, and nutrient-rich meals. This integration would help users maintain both fitness and overall wellness more effectively.

3. Cloud Synchronization and Authentication

Future development may include Firebase Authentication and cloud database integration for secure account management and real-time data synchronization. This would allow users to access their workout history, progress data, and preferences across multiple devices while ensuring safe backup and recovery of personal information.

4. Wearable Device Integration

BloomBalance can be integrated with wearable devices such as smartwatches and fitness

bands to monitor real-time health metrics including heart rate, sleep quality, calories burned, and physical activity levels. This would improve workout accuracy and enable better health analysis and fitness tracking.

5. Advanced Cycle Tracking and Prediction

The application can be enhanced with advanced menstrual cycle prediction features using historical cycle data and health patterns. More accurate cycle forecasting and symptom prediction would allow users to plan workouts and recovery periods more efficiently.

6. Community and Social Features

Future versions may include community-based features such as fitness challenges, support groups, progress sharing, and social interaction. These features can improve user motivation, consistency, and engagement by creating a supportive fitness environment.

7. Enhanced Analytics and Insights

The analytics system can be improved with detailed reports, long-term performance analysis, and personalized wellness insights. Users would be able to better understand their progress, recovery trends, and fitness consistency through advanced visualizations and data analysis.

8. Virtual Wellness Assistant

A chatbot or virtual wellness assistant can also be integrated into the application to provide real-time fitness guidance, symptom support, workout suggestions, and motivational interactions. This feature would create a more interactive and user-friendly experience for users.

REFERENCES

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